Improved robustness of deep learning models through posterior agreement based model selection

Master Thesis

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Experimental pathway

Posterior Agreement has been proposed as a theoretically-grounded alternative for model robustness assessment in covariate shift settings.

1. Characterization of the sources of randomness that are relevant in the context of image classification tasks.
2. Posterior Agreement as a robustness metric.
3. Robustness assessment in the adversarial setting.
4. Robustness assessment in the out-of-distribution setting.
5. Model selection with early-stopping.

The robustness challenge – Introduction

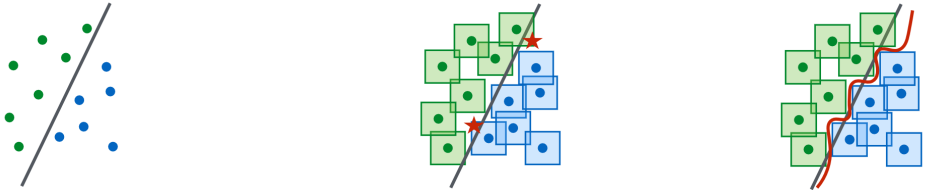
Goal: Maintain predictive power under expected variations in the data.

(a) Original (b) Sampling (c) Adversarial (d) Distribution

Figure: Illustrative example of the three expected sources of randomness in the context of image classification.

The robustness challenge – Challenges

- Lack of understanding of the inductive bias.
- Generalization-complexity trade-off.
- Robust vs non-robust features.



(a) Linsep points (b) Standard training (c) Adversarial training (ℓ_∞)

Figure: Standard vs adversarial training decision boundaries.

Learning framework – Introduction

Goal: Learn a target function $f^* : \mathcal{X} \longmapsto \mathcal{Y}$ by means of an approximated function $f \in \mathcal{F}$ using a finite set of observations.

- Images $x \in \mathcal{X}$ belonging to $K = |\mathcal{Y}|$ classes.
- f^* encodes a causal structure driving the data generation process.
- $\mathcal{F}$ is composed of parametrized classifiers.

Learning framework – Samples

Definition (*Sample*):

- Let X be a random variable associated with a sampling experiment in the input space, thus defining a measure of probability P_X with support $\mathcal{X}$.
- Let $\boldsymbol{X} = (X_1, ..., X_N)$ be a simple random sample of X with size N.

A sample is a realization $\boldsymbol{x} \sim \boldsymbol{X}$.

A supervised dataset is constructed from $\boldsymbol{x}$ as:

$$D = \{(x_n, f^*(x_n))\}_{n \in [N]}.$$

Learning framework – Model

Definition (*Classifier*): Let $K \in \mathbb{N} < \infty$ be the cardinality of $\mathcal{Y}$.

$$\begin{array}{rcccccc} f : \mathcal{X} & \longmapsto & \mathbb{R}^d & \longmapsto & \mathbb{R}^K & \longmapsto & \mathcal{Y} = \{1, \ldots, K\} \\ x & \longmapsto & z & \longmapsto & \boldsymbol{F}(z) & \longmapsto & \hat{y} = \arg\max_k F_k(z) \end{array}$$

Under a NN parametrization Γ:

$$\begin{aligned} f : \mathcal{X} \times \Gamma &\longmapsto \mathcal{Y} = \{1, \ldots, K\} \\ (x, \gamma) &\longmapsto f(x; \gamma) = \hat{y}, \end{aligned}$$

Learning framework – Algorithm

Definition (*K-class classification problem*):

- Let L be the cross-entropy loss function.

$$L(y) = -\log F_y(z;\gamma)$$

- Let $\hat{R}(f)$ be the empirical risk.

$$\hat{R}(f) = \frac{1}{N}\sum_{n=1}^{N} L(f(x_n), f^\star(x_n))$$

$$\boxed{\gamma^* = \arg\min_{\gamma\in\Gamma} -\frac{1}{N}\sum_{n=1}^{N} \log F_{y_n}(x_n;\gamma) + \lambda\Omega(\gamma)}$$

Posterior Agreement

Definition (*Hypothesis class*) A data science algorithm learns a function f implementing the following mapping:

$$\begin{aligned} f : \boldsymbol{X} &\longmapsto \Theta \\ \boldsymbol{x} &\longmapsto (f(x_1), \ldots, f(x_N)) = \theta. \end{aligned}$$

Definition (*Posterior*) The posterior $\mathbf{P}^f \in \mathfrak{P}^f$ establishes the stochastic relation between experiment realizations and hypotheses.

$$\begin{aligned} \mathbf{P}^f : \boldsymbol{X} \times \Theta &\longmapsto \mathbb{R} \\ (\boldsymbol{x}, \theta) &\longmapsto \mathbf{P}^f(\theta \mid \boldsymbol{x}). \end{aligned}$$

Posterior Agreement

Definition (*Generalization error*):

- Let $\boldsymbol{x}'$ and $\boldsymbol{x}''$ be realizations of $\boldsymbol{X}$.
- Let Θ be the hypothesis class represented by f given $\boldsymbol{X}$.
- Let $-\log \mathbf{P}^f(\cdot)$ be the description length of the posterior.

The generalization error is defined as the out-of-sample description length:

$$\boxed{\mathcal{G}_{\mathcal{X}} = \mathbb{E}_{\boldsymbol{x}',\boldsymbol{x}''}\mathbb{E}_{\mathbf{P}^f(\theta|\boldsymbol{x}')}\left[-\log \frac{\mathbf{P}^f(\theta|\boldsymbol{x}'')}{\Pi^f(\theta)}\right].}$$

Intuitively, a low generalization error is obtained when good quality hypothesis on $\boldsymbol{x}''$ are likely to be drawn from $\boldsymbol{x}'$.

Lemma (*Posterior agreement*): The generalization error $\mathcal{G}_{\mathcal{X}}$ is non-negative and has a lower bound $-\mathcal{J}$.

$$
\begin{aligned}
\mathcal{G}_{\mathcal{X}} &\geq \mathbb{E}_{\boldsymbol{x}',\boldsymbol{x}''}\left[-\log\left(\mathbb{E}_{\mathbf{P}^f(\theta|\boldsymbol{x}')}\frac{\mathbf{P}^f\left(\theta \mid \boldsymbol{x}''\right)}{\Pi^f(\theta)}\right)\right] \\
&= \boxed{\mathbb{E}_{\boldsymbol{x}',\boldsymbol{x}''}\left[-\log\left(\sum_{\theta\in\Theta}\frac{\mathbf{P}^f\left(\theta \mid \boldsymbol{x}'\right)\mathbf{P}^f\left(\theta \mid \boldsymbol{x}''\right)}{\Pi^f(\theta)}\right)\right] = -\mathcal{J}} \\
&\geq -\log\left(\mathbb{E}_{\boldsymbol{x}',\boldsymbol{x}''}\mathbb{E}_{\mathbf{P}^f(\theta|\boldsymbol{x}')}\mathbb{E}_{\mathbf{P}^f(\theta|\boldsymbol{x}'')}\frac{\mathbf{P}^f\left(\theta \mid \boldsymbol{x}''\right)}{\Pi^f(\theta)}\right) = 0,
\end{aligned}
$$

where Jensen's inequality has been applied twice to the convex function $-\log$.

Proposition: Posterior agreement based model selection criterion.

$$\sup_{\mathcal{F}} \mathcal{J}$$
$$\text{s.t. } \mathsf{KL}(\mathbf{\Pi}^f(\theta) \parallel |\Theta|^{-1}) \leq \xi,$$

where $\xi \in \mathbb{R}$ represents a small allowed deviation from uniformity in the prior.

Theorem (*Maximum posterior agreement*): The optimal $\mathbf{P}_*^f$ maximizing the posterior agreement criterion defines a lower bound in the generalization error $\mathcal{G}_{\mathcal{X}}$:

$$\inf_{\mathcal{F}} \mathcal{G}_{\mathcal{X}} \geq -\sup_{\mathcal{F}} \mathcal{J}.$$

Proposition (*Posterior Agreement kernel*): With no prior information about Θ, the posterior agreement kernel for supervised K-class classification tasks has the following expression:

$$\boxed{\mathrm{PA}\left(\boldsymbol{x}', \boldsymbol{x}''; \beta\right) = \frac{1}{N} \sum_{n=1}^{N} \log \left\{ |\Theta| \sum_{k=1}^{K} \mathbf{P}^{c}\left(k \mid x_n'\right) \mathbf{P}^{c}\left(k \mid x_n''\right) \right\},}$$

where $\mathbf{P}^{c}(j \mid x_n)$ can be shown to be

$$\boxed{\mathbf{P}^{c}\left(k \mid x_n\right) = \frac{\exp\left(\beta F_k(x_n)\right)}{\sum_{q=1}^{K} \exp\left(\beta F_q(x_n)\right)}.}$$

Theorem (*Properties of the PA kernel*): The posterior agreement kernel for K-classification problems $\mathrm{PA}\left(\boldsymbol{x}', \boldsymbol{x}''; \beta\right)$ has the following properties $\forall \boldsymbol{x}', \boldsymbol{x}'' \sim \boldsymbol{X}$ and $\beta \in \mathbb{R}^+$.

- P1 (Boundedness) $\mathrm{PA}\left(\boldsymbol{x}', \boldsymbol{x}''; \beta\right) \leq N \log K$. Because $\log_2 K$ bits are needed to encode a uniform distribution over the classes for each observation.
- P2 (Symmetry) $\mathrm{PA}\left(\boldsymbol{x}', \boldsymbol{x}''; \beta\right) = \mathrm{PA}\left(\boldsymbol{x}'', \boldsymbol{x}'; \beta\right)$. Because randomness instantiations are not observable.
- P3 (Concavity) $\mathrm{PA}\left(\boldsymbol{x}', \boldsymbol{x}''; \beta\right)$ is a concave function of $\beta \in \mathbb{R}^+$. Then the kernel optimization problem will have a unique solution.

In particular, the maximum value $\mathrm{PA} \equiv \mathrm{PA}\left(\boldsymbol{x}', \boldsymbol{x}''; \beta^*\right)$ is bounded by the situations $\beta^* \longrightarrow 0$ and $\beta^* \longrightarrow \infty$. Then:

$$-N \log K \leq \mathrm{PA} \leq 0$$

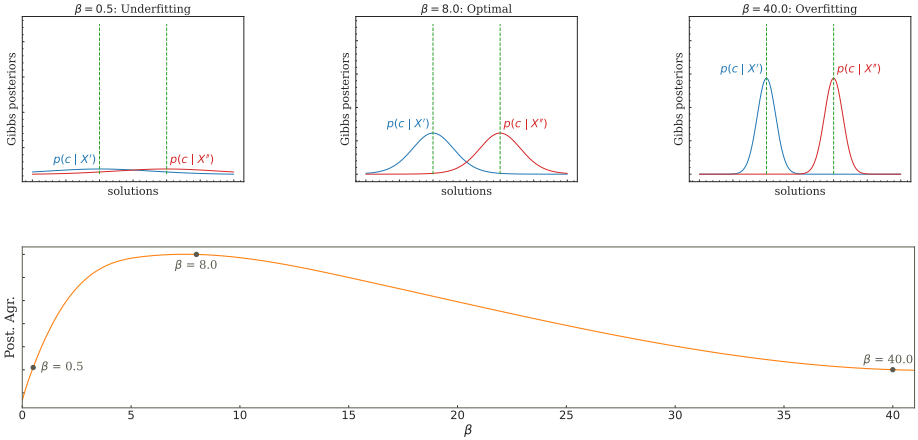


Figure: Illustration of the optimization over the inverse temperature parameter β. Posterior Agreement is maximum at a value β^* in which hypothesis selected from the posterior over $\theta \mid \boldsymbol{x}'$ are assigned a high probability by the posterior over $\theta \mid \boldsymbol{x}''$.

Experimental setup – Covariate shift

Definition (*Covariate shift*): Let $\boldsymbol{x}'$ and $\boldsymbol{x}''$ be two samples of $\boldsymbol{X} \overset{\text{iid}}{\sim} X$.

$$\mathbf{P}_{\boldsymbol{x}'} \not\sim \mathbf{P}_{\boldsymbol{x}''}.$$

Definition (*Distribution shift*): Let X' and X'' be two random variables associated to different sampling experiments in $\mathcal{X}$ such that $P_{X'} \not\sim P_{X''}$.

$$\boldsymbol{x}' \sim \boldsymbol{X}' \overset{\text{iid}}{\sim} X' \text{ and } \boldsymbol{x}'' \sim \boldsymbol{X}'' \overset{\text{iid}}{\sim} X''$$

Definition (*Adversarial shift*): Let $\boldsymbol{x}' \sim \boldsymbol{X}$ be a sample drawn from experiment X. Let $\boldsymbol{\Delta}$ be a perturbation over the sample space.

$$\boldsymbol{x}'' = \boldsymbol{x}' + \boldsymbol{\Delta},$$

Experimental setup – Robustness measurement

Proposition. A robustness metric should possess the following properties:

P1 (Non-increasing) The metric should be non-increasing with respect to the response of the model under increasing levels of covariate shift.

P2 (Independent discriminability) The metric should discriminate models exclusively by their generalization capabilities against covariate shift. For instance, the metric should be independent of the task performance.

Example I. Consider the following classifiers in a balanced dataset:

Classifier	Performance	Robustness
Perfect	1.0	Max.
Constant	$1/K$	Max.
Random	$1/K$	Min.

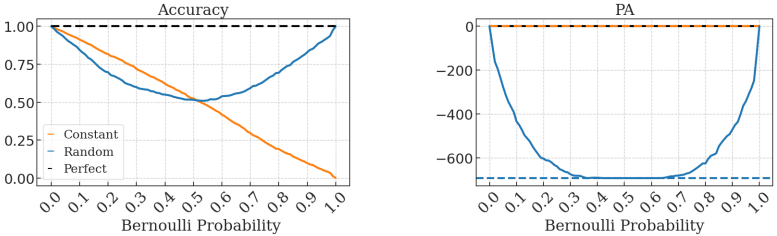


Figure: Evolution of PA and accuracy for constant, perfect and random classifiers across different values of $p \in [0, 1]$. Accuracy does not comply with the desired properties of a robustness metric and provides an inconsistent assessment that is exclusively driven by task performance. In contrast, PA continuously discriminates robust from unrobust classifiers for $p \in (0, 1)$ and reaches its minimum $-N \log 2$ (blue dashed line) for the random classifier when $p \in (0.3, 0.7)$.

Example II. Consider a sentiment classifier for IMDB dataset. $\boldsymbol{x}'$ is a sample of original reviews, and $\boldsymbol{x}''$ contains a manipulated version of these.

- Levenshtein: Addition, removal or substitution of characters.
- Amplification: Addition of reinforcing adjectives.
- Contradiction: Addition of contradicting adjectives.

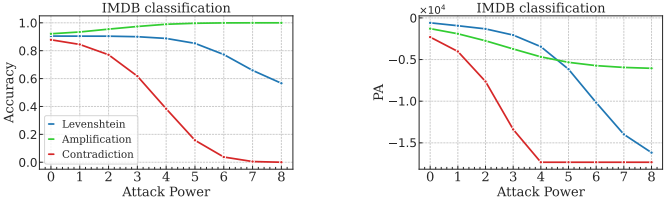


Figure: PA and accuracy for the IMDB sentiment classification task under random and adversarial perturbations. The attack power is defined as 2^W, being W the number of modifications performed.

Adversarial setting – Attacks

Definition (*Perturbation*) Let $\mathbf{B}_p^\epsilon(x)$ be the ℓ_p-norm ball of radius ϵ centered at observation x. A perturbation Δ is defined as

$$\Delta \in \mathbb{R}^d \text{ s.t. } x + \Delta \in \mathbf{B}_p^\epsilon(x),$$

Attack (*PGD*): Projected gradient descent.

$$x^{s+1} = \Pi_{\mathbf{B}_p^\epsilon(x)}\left(x^s + \Delta\right); \quad \Delta = \epsilon_p \operatorname{sign}(\nabla_{x'}\mathcal{L}(x', y; \gamma))$$

Attack (*FMN*): Fast minimum norm.

$$\begin{aligned} \Delta^\star = \arg\min_{\Delta} \; & ||\Delta||_p \\ \text{s.t. } & F_y(x;\gamma) - \max_{k \neq y} F_k(x;\gamma) < 0, \\ & x + \Delta \in \mathbf{B}_p^\epsilon(x). \end{aligned}$$

Adversarial setting

Experimental setup. CIFAR10 [6] is a balanced dataset containing 60.000 colored 32×32 pixel images belonging to 10 different classes. We will consider a pre-trained WideResNet-28-10 [11] as a baseline, undefended model and compare it to some state-of-the-art robust ResNet50 [5] models provided by the RobustBench [3] library under PGD [7] and FMN [8] attacks, both run for 1000 steps. The defenses applied are those proposed by **Engstrom et al.** [4], **Athalye et al.** [2], **Wong et al.** [10], **Addepalli et al.** [1] and **Wang et al.** [9].

(a) Original

(b) PGD, $\ell_\infty = 36/255$

(c) FMN

Figure: Original and adversarially-perturbed CIFAR10 observation of class `horse`. Both perturbations succeed at misleading an undefended, pre-trained WideResNet-28-10.

Adversarial setting – PGD

Experiment 1. A pre-trained, undefended WideResNet-28-10 and five RobustBench defended models are subject to a 1000 step PGD attack with $\ell_\infty = 8/255$.

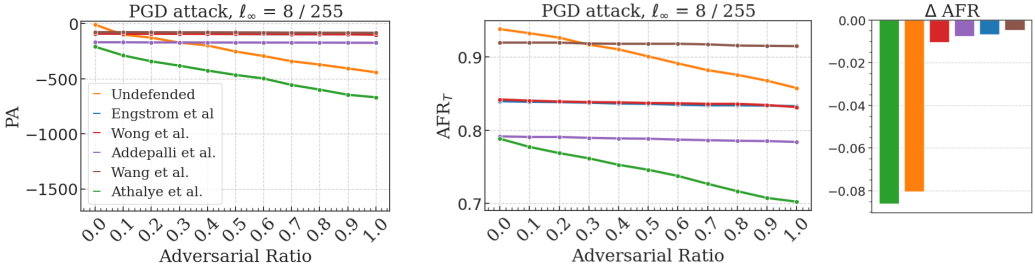


Figure: PA, $\mathrm{AFR}_\mathcal{T}$ and the AFR variation against increasing adversarial ratio $\alpha \in [0, 1]$.

Out-of-distribution setting

- **IRM** (domain alignment): Regularization term that pushes towards the minimization of the dissimilarity of feature representations originated from different source environments.
- **LISA** (data augmentation): Artificial observations are generated in between labels or in between domains

Model

Start with some bullet points:

- First, ...
- Second, ...

All important mathematical numbers in one formula

$$\exp\{-i\pi\} + 1 = 0.$$

More text, ...

Definition

Content...

Analysis

Content...

Theorem

Content...

Conclusion

Content...

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